A **Survey** on Anomaly detection in Evolving Data: [with Application to **Forest Fire** Risk **Prediction**]

M Salehi, L Rashidi - ACM SIGKDD Explorations Newsletter, 2018 - dl.acm.org

… that **forest fire** risk needs to consider both **forest fire** behaviour … temporal factors controlling **forest fire spread**, including fuel and … **maps** derived from gis modeling and direct **simulation** …

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Review of **wildfire management** techniques—Part I: Causes, prevention, detection, suppression, and data analytics

S Jazebi, F De Leon, A Nelson - IEEE Transactions on Power …, 2019 - ieeexplore.ieee.org

… frames over time for remote monitoring of **wildfire spread**. In [45], a … **forest fire** with low consumption of energy. The fire **spread model** in [46] can produce fire **propagation maps** …

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Using cellular automata to simulate **wildfire propagation** and to assist in fire **management**

JG Freire, CC DaCamara - Natural hazards and earth system …, 2019 - nhess.copernicus.org

… Figure 1(a) **Map** of Portugal with the location of the Tavira **wildfire**, where orange represents the burned scar and the black frame indicates the study area used in the simulations. (b, c) …

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Advancements in **forest fire** prevention: A comprehensive **survey**

F Carta, C Zidda, M Putzu, D Loru, M Anedda, D Giusto - Sensors, 2023 - mdpi.com

… [10] conducted a study on **wildfire propagation** in Portugal, … in an accurate fire **susceptibility map**. Eight variables related to … [118] introduced a **wildfire spread model** implementing an …

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Global pyrogeography: the current and future distribution of **wildfire**

MA Krawchuk, MA Moritz, MA Parisien, J Van Dorn… - PloS one, 2009 - journals.plos.org

… -climate change **dynamics** and conservation planning. … occurrence models from the GFDL CM2.1 historical **simulation** … near-term shifts, we **mapped** the distributions of fire invasion and …

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Next day **wildfire spread**: A machine learning dataset to predict **wildfire** spreading from remote-sensing data

F Huot, RL Hu, N Goyal, T Sankar… - IEEE Transactions on …, 2022 - ieeexplore.ieee.org

… Day **Wildfire Spread**" dataset, we train a deep learning **model** … input image and the fire mask as a segmentation **map**. … A **review** of machine learning applications in **wildfire** science …

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Land use planning and **wildfire**: development policies influence future probability of housing loss

AD Syphard, A Bar Massada, V Butsic, JE Keeley - PloS one, 2013 - journals.plos.org

… changing **dynamics** in the development **map** of the county. … We first developed the **model** using **mapped** explanatory … projected the **model** conditioned upon current conditions onto **maps** …

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FireCast: Leveraging Deep Learning to Predict **Wildfire Spread**.

D Radke, A Hessler, D Ellsworth - IJCAI, 2019 - ijcai.org

… compared to a random **prediction model** and a commonly used **wildfire spread model**, Farsite, … Getting ahead of the **wildfire** problem: quantifying and **mapping management** challenges …

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Fire behaviour and smoke modelling: **model** improvement and measurement needs for next-generation smoke research and forecasting systems

Y Liu, A Kochanski, KR Baker, W Mell… - … of **wildland fire**, 2019 - CSIRO Publishing

… and safety concerns associated with **wildland fire** emissions. This **review** paper presents simulations and … The fire's rate of **spread** is obtained from the Rothermel **model**. In its simplest …

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Parameter flexible **wildfire prediction** using machine learning techniques: Forward and inverse modelling

S Cheng, Y Jin, SP Harrison, C Quilodrán-Casas… - Remote Sensing, 2022 - mdpi.com

… **propagation** using estimated/adjusted parameters. After ROM, we aim to learn the **dynamics** … parameters and the encoded observations y t are **mapped** through ML functions, the direct …

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Human-caused fire occurrence modelling in perspective: a **review**

S Costafreda-Aumedes, C Comas… - … of **Wildland Fire**, 2017 - CSIRO Publishing

… , and is a process modelled separately from fire **spread**, which may or may not take place … causality, improving global **wildfire** databases, novel **model** development, some normalisation …

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A **simulation** of probabilistic **wildfire** risk components for the continental United States

MA Finney, CW McHugh, IC Grenfell, KL Riley… - … Research and Risk …, 2011 - Springer

… To make the fire growth **simulation** efficient, fire **behavior** for … the centroid for the Public Land **Survey** System (PLSS) section in … developed areas that are **mapped** as having no flammable …

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Wildfire management in Canada: **Review**, challenges and opportunities

C Tymstra, BJ Stocks, X Cai, MD Flannigan - Progress in Disaster Science, 2020 - Elsevier

… **management** agencies to allow more **wildfire** on the landscape by implementing a risk-based … future path for **wildfire management** in Canada. To co-exist with **wildfire**, agencies in …

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Modeling vorticity-driven **wildfire behavior** using near-field techniques

JJ Sharples, JE Hilton - Frontiers in Mechanical Engineering, 2020 - frontiersin.org

… , they provide fire **management** end-users with a way to **map** the likely evolution of an active … **simulation** using the pyrogenic vector potential coupled to a **wildfire spread simulation** is …

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Using soil moisture information to better understand and predict **wildfire** danger: a **review** of recent developments and outstanding questions

ES Krueger, MR Levi, KO Achieng, JD Bolten… - … journal of **wildland fire**, 2023 - connectsci.au

… temporal resolutions to inform **wildfire management**. The International … of the **wildfire** (eg rate of **spread** and fire intensity). … for **wildland fire** managers in Oklahoma that produces **maps** of …

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An overview of mountain meteorological effects relevant to fire behaviour and **bushfire** risk

JJ Sharples - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… The likelihood of growth of a **bushfire** is typically derived using some form of fire **spread model** that … risk in the Australian high country, it is prudent to begin with a **review** of the available …

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Review of **wildfire** detection using social media

V Slavkovikj, S Verstockt, S Van Hoecke… - Fire safety journal, 2014 - Elsevier

… in the context of **wildfire propagation**. The authors evaluated … **model map** and used as a ground truth of the **model forecast**. … oil **propagation model** by correlating the **model** predictions …

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Deep neural networks for global **wildfire susceptibility** modelling

G Zhang, M Wang, K Liu - Ecological Indicators, 2021 - Elsevier

… training, validation and testing, seasonal global **susceptibility maps** are constructed, … **prediction** performance of the models. The results show that the contextual-based CNN-2D **model** …

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Efficient **simulation** of **wildfire spread** on an irregular grid

P Johnston, J Kelso, GJ Milne - … Journal of **Wildland Fire**, 2008 - CSIRO Publishing

… **wildfire simulation** techniques because computing resources are not used in repeatedly computing small updates to parts of the fire whose **dynamics** … At any point in time, a **map** can be …

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… data-driven simulations of **wildfire spread**–Part I: Reduced-cost Ensemble Kalman Filter based on a Polynomial Chaos surrogate **model** for parameter estimation

MC Rochoux, S Ricci, D Lucor… - Natural Hazards and …, 2014 - nhess.copernicus.org

… using a reference solution of the FIREFLY fire **spread model** (called the true evolution) that is … ture **maps** and using a definition of the fire front as the 600 K temperature contour line. …

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